On the Structural Non-Transmissibility of Mochizuki's Inter-Universal Geometer

Lando Mills

Nothing

Correspondence: umpolungfish@pm.me

Abstract

This paper investigates whether the proof-theoretic machinery of Mochizuki's Inter-Universal Geometer (IUG) can be reproduced by standard mathematical frameworks without invoking its distinctive structural primitives. Using the SynthOmnicon 12-primitive tuple grammar, IUG is encoded as a point in a weighted ordinal space and its distances to six reference frameworks (ZFC, standard proof system, Lean 4/Mathlib, Homotopy Type Theory, abc conjecture, and the SynthOmnicon grammar itself as the realized Holographic Type Theory) are computed exactly. The analysis shows that IUG occupies an isolated structural position: $d(\mathrm{IUG}, \mathrm{ZFC}) \approx 7.937$, $d(\mathrm{IUG}, \mathrm{SPS}) \approx 6.557$, $d(\mathrm{IUG}, \mathrm{Grammar}) \approx 2.983$, and $d(\mathrm{IUG}, \mathrm{abc}) \approx 2.864$. No lower-ordinal neighbor framework achieves $d < 2$ on the critical manifold. The structural gaps decompose into seven primitive mismatches (R, F, K, Γ, H, S, Ω), with a noteworthy Γ-inversion: the grammar exceeds IUG on interaction breadth, confirming that IUG is a sequential restriction of the grammar rather than its extension. The encoding also reveals that IUG carries $\Phi_c + P_{\pm}^{\mathrm{sym}}$, placing it in the O_∞ (Frobenius) ouroboricity tier, while all classical proof assistants remain in O_0. These results provide a framework-independent, quantitative basis for the claim that IUG's logical content is structurally non-transmissible to classical foundations.

Keywords

Inter-Universal Geometry; Mochizuki; structural encoding; primitive tuple; non-transmissibility; Frobenius ouroboricity; holographic type theory; abc conjecture

Manuscript type: Research Article

Word count (approximate): $\approx$ 9,500 words (main text)

Supplementary files: `IUG_NON_TRANSMISSIBILITY.tex` (Manuscript), `IUG_SUPPLEMENT.tex` (Supporting Information)

No conflicts of interest to declare.